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Integrated circuit device package

Abstract

An example apparatus includes: an integrated circuit including a first surface and terminals; a package including: a housing around the integrated circuit, the housing exposing the first surface; and an electrical interconnect including a second surface and an opening, the second surface electrically coupled to the terminals, the second surface mechanically coupled to the housing, the opening configured to expose the first surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This patent application claims the benefit of and priority to U.S. Provisional Patent Application Ser. No. 63/087,944 filed Oct. 6, 2020, which application is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) This description relates generally to integrated circuits, and more particularly to methods and apparatus for an integrated circuit device package.

BACKGROUND

(2) Integrated circuits (ICs) may receive electrical inputs and generate electrical outputs. An optical IC may receive an electrical input and generate an optical output. In an example, optical ICs may receive an optical input and generate an electrical output. Thus, optical IC packages accommodate both an electrical interface (to support the electrical input and/or the electrical output) and an optical interface (to support the optical input and/or the optical output). An optical interface is a portion of the IC package designed to allow optical signals (e.g., light) to enter and/or exit the IC package. Conventionally, accommodating for the optical interface increases the optical IC package size due to mechanical alignment features needed to attach the optical IC to a printed circuit board (PCB).

SUMMARY

(3) For methods and apparatus to package an integrated circuit device, an example apparatus includes an integrated circuit including a first surface and terminals; a package including: a housing around the integrated circuit, the housing exposing the first surface; and an electrical interconnect including a second surface and an opening, the second surface electrically coupled to the terminals,

the second surface mechanically coupled to the housing, the opening configured to expose the first surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an isometric view of a first example integrated circuit.
- (2) FIG. 2 illustrates an isometric view of a first example housing constructed in accordance with the teachings disclosed herein.
- (3) FIG. 3 illustrates an isometric view of a first example assembly including the first integrated circuit of FIG. 1 and the first housing of FIG. 2.
- (4) FIG. 4 illustrates a first isometric view of a second example assembly including the first assembly of FIG. 3 and a first example electrical interconnect.
- (5) FIG. 5 illustrates a second isometric view of the second example assembly of FIG. 4.
- (6) FIG. 6 illustrates a third isometric view of the second example assembly of FIG. 4.
- (7) FIG. 7 illustrates a bottom view of the first electrical interconnect of FIGS. 4-6.
- (8) FIG. 8 illustrates a bottom view of a second example electrical interconnect.
- (9) FIG. 9 illustrates a cross-sectional view of an example apparatus to assemble the second assembly of FIG. 4.
- (10) FIG. 10 illustrates a bottom view of a third example assembly including the second assembly of FIG. 4 and a second example integrated circuit.
- (11) FIG. 11 illustrates a first cross-sectional view of the second integrated circuit of FIG. 10 along line 14-14 of FIG. 10.
- (12) FIG. 12 illustrates a second cross-sectional view of the second integrated circuit of FIG. 10 along line 14-14 of FIG. 10 including an example adhesive.
- (13) FIG. 13 illustrates a third cross-sectional view of the third assembly of FIG. 10 along line 14-14 of FIG. 10.
- (14) FIG. 14 illustrates a fourth cross-sectional view of the third assembly of FIG. 10 taken along line 14-14 of FIG. 10.
- (15) FIG. 15 is a flowchart representative of an example method to manufacture the third assembly of FIG. 10.
- (16) FIG. 16 illustrates a top view of a fourth example assembly including the first assembly of FIG. 3 and a third example electrical interconnect.
- (17) FIG. 17 illustrates an isometric view of the fourth assembly of FIG. 16.
- (18) FIG. 18 illustrates a side view of a fifth example assembly including the fourth assembly of FIGS. 16 and 17, and a fourth example electrical interconnect.
- (19) FIG. 19 is a flowchart representative of an example method to manufacture the fifth assembly of FIG. 18.
- (20) FIG. 20 illustrates a side view of a sixth example assembly including the first integrated circuit of FIG. 1, a plurality of electrical interconnects, and a plurality of stand-offs.
- (21) FIG. 21 is a flowchart representative of an example method to manufacture the sixth assembly of FIG. 20.
- (22) FIG. 22 is a side view of a seventh example assembly.
- (23) FIG. 23 is a side view of an eighth example assembly.
- (24) FIG. 24 is a side view of a ninth example assembly.
- (25) FIG. 25 is a side view of a tenth example assembly including a second housing.
- (26) FIG. 26 is a side view of an eleventh example assembly.
- (27) FIG. 27 is a side view of a twelfth example assembly.
- (28) FIG. 28 is a side view of a third example housing.

(29) FIG. 29 is a side view of a thirteenth example assembly including the third housing of FIG. 28.
(30) FIG. 30 is a side view of a fourteenth example assembly including the thirteenth assembly of FIG. 29.

(31) FIG. 31 is a side view of a fifteenth example assembly including the thirteenth assembly of FIG. 29.

DETAILED DESCRIPTION

(32) The drawings are not necessarily to scale. Generally, the same reference numbers in the drawing(s) and this description refer to the same or like parts. Although the drawings show layers and regions with clean lines and boundaries, some or all of these lines and/or boundaries may be idealized. In reality, the boundaries and/or lines may be unobservable, blended and/or irregular.

(33) Electrical ICs may receive electrical inputs and generate electrical outputs. An optical IC may receive an electrical input and generate an optical output. In an example, optical ICs may receive an optical input and generate an electrical output. Example optical ICs may include spatial light modulators, digital micromirror devices, optical sensors, optical sources, an optical device, and optical processors, etc. An optical IC may be referred to as an optical device. For example, a digital micromirror device (DMD) is an optical IC that receives an electrical input, an optical input (e.g., light), and generates an optical output (e.g., an image). A DMD may be implemented to receive digital video data as an electrical input and generate an image as an optical output. The optical output of the DMD is based on the way the electrical interface controls the DMD to reflect light from the optical input, such that images may be displayed. In some such applications, the optical IC packaging includes an opening, such that the optical output may be viewed.

(34) Optical IC packages accommodate both an electrical interface (to support the electrical input and/or the electrical output) and an optical interface (to support the optical input and/or the optical output). An optical interface is a portion of the IC package designed to allow optical signals (e.g., light) to enter and/or exit the IC. An electrical interface is a portion of the IC package designed to be electrically coupled to additional circuitry (e.g., a PCB, a processing circuit, a receiver, etc.). In some applications, such as the DMD, the electrical interface is designed to be electrically coupled to another integrated circuit in such a way that results in the optical interface being aligned, such that the optical input and/or output operate as designed. For example, a DMD may align the optical interface on the optical IC before aligning the electrical interface on the PCB.

(35) Accommodating the optical interface typically increases the optical IC package size by including features to mechanically align the optical IC to a PCB. Some optical IC applications allow mechanical alignment features to be included on the PCB. Including the mechanical alignment features on the PCB allows for a smaller optical IC package. Some electrical IC applications allow the electrical IC die to attach directly to the PCB. Some optical ICs use wire bonding to electrically couple to a PCB.

(36) The methods and apparatus to package an integrated circuit device described herein include an alignment method, such as a mechanical alignment method, for an electrical interface and/or an optical interface. The mechanical alignment method may be configured to mechanically align the electrical interface and optical interface of an optical IC. The mechanical alignment method implements an intermediate structural carrier (e.g., a housing) to align an electrical coupling of the optical IC and an external PCB. The intermediate structural carrier may be implemented to align the optical interface of the optical IC as a result of the electrical coupling. The intermediate structural carrier enables an electrical interconnect to easily and durably be mechanically and/or electrically coupled to the optical IC. The electrical interconnect enables the optical IC to easily and durably be electrically coupled to an IC. The intermediate structural carrier and electrical interconnect enable an optical IC to be durably electrically and/or mechanically coupled to an IC, such that both the optical interface and the electrical interface are aligned.

(37) FIG. 1 illustrates an isometric view of a first example integrated circuit 100. As shown in the illustrated example, the first integrated circuit 100 includes an example optical IC 110. The optical

IC **110** may be configured to include circuitry supporting electrical and optical, input and/or output. The optical IC **110** includes a first example plurality of terminals **120**, a second plurality of terminals **130**, an optical interface **140**, and a surface **150**. Additionally, the optical IC **110** may internally include circuitry (e.g., memory, processing logic, etc.) and mechanical components (e.g., micromirrors, light sources, etc.) to support the operation of the first integrated circuit **100**. The plurality of terminals **120** and **130** form all or part of an electrical interface. In the example of FIG. **1**, the first integrated circuit **100** is the optical IC **110** mechanically coupled to the optical interface **140**, such that the optical interface **140** and the electrical interface (including the first plurality of terminals **120** and the second plurality of terminals **130**) are included on the surface **150**.

(38) FIG. **2** illustrates an isometric view of a first example housing **200**. As shown in the illustrated example, the first housing **200** includes a surface **210**, a first example opening **220** and a first example edge **230**. In the example of FIG. **2**, the first opening **220** includes dimensions similar to (in some cases exactly the same as) the dimensions of the optical IC **110** of FIG. **1**. The dimensions of the first edge **230** are based on the difference between the dimensions of the surface **210** and the dimensions of the first opening **220**. The first edge **230** is configured to extend around the first opening **220** to the edge of the surface **210**. Alternatively, the first edge **230** may be configured to extend around a portion of the first opening **220** or implemented with a plurality of edges. The first housing **200** may be manufactured using an electrical insulator, such as an epoxy material.

(39) FIG. **3** illustrates an isometric view of a first example assembly **300** including the first integrated circuit **100** of FIG. **1** and the first housing **200** of FIG. **2**. In the example of FIG. **3**, the first integrated circuit **100** is shown inserted into the first opening **220** of the first housing **200**, such that the surface **150** of the first integrated circuit **100** is aligned (e.g., substantially flush) with the surface **210** of the first housing **200**. The first plurality of terminals **120** and the second plurality of terminals **130** are configured to protrude out of the surface **150** of the first integrated circuit **100**. Alternatively, the terminals may be flush with the surface **150** of the first integrated circuit **100**. Advantageously, the first edge **230** is configured increase the surface size of the surface **150**, such that the potential contact area during manufacturing is greater.

(40) FIG. **4** illustrates a first isometric view of a second example assembly **400** including the first assembly **300** of FIG. **3** and a first example electrical interconnect **410**. The second assembly **400** includes the first integrated circuit **100** of FIGS. **1** and **3**, the first housing **200** of FIGS. **2** and **3**, and the first electrical interconnect **410**. The first electrical interconnect may be manufactured as a printed circuit board (PCB) including a single semiconductor die or on multiple die. Alternatively, the first electrical interconnect **410** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.) and insulator, designed to electrically couple a plurality of terminals to another plurality of terminals. In the example of FIG. **4**, the first electrical interconnect **410** includes a second example opening **420**, an example surface **430**, and a third example plurality of terminals **440**, a fourth example plurality of terminals **450**, and a fifth example plurality of terminals **460**. The second opening **420** is sized to enable the optical interface **140** of the first integrated circuit **100** to be exposed. The second opening **420** may additionally be configured to include additional space that may be used as a mechanical guide for coupling the first electrical interconnect **410** to the first assembly **300** of FIG. **3**.

(41) In the example of FIG. **4**, the surface **430** includes the third plurality of terminals **440**, the fourth plurality of terminals **450**, and the fifth plurality of terminals **460**. The third plurality of terminals **440** are configured to protrude from the surface **430** of the first electrical interconnect. Alternatively, the third plurality of terminals **440** may be flush with the surface **430**. The fourth plurality of terminals **450** and fifth plurality of terminals **460** may be electrically coupled to the third plurality of terminals **440** through conductors routed through the first electrical interconnect **410**. For example, the plurality of terminals **450** and **460** may be configured to be electrically coupled to the third plurality of terminals **440**, such that each terminal in the third plurality of terminals corresponds to a terminal in the plurality of terminals **450** and **460**. The fourth plurality

of terminals **450** may be electrically coupled to the first plurality of terminals **120** of FIG. **1**. The fifth plurality of terminals **460** may be electrically coupled to the second plurality of terminals **130** of FIG. **1**. In the example of FIG. **4**, the surface **150** of the first integrated circuit **100** and the surface **210** of the first housing **200** (not pictured in FIG. **4** for clarity) are configured to be mechanically coupled to the surface **430** of the first electrical interconnect **410**, such that the first electrical interconnect **410** covers the surface **210** of the first housing **200**. Alternatively, the first electrical interconnect **410** may be configured to cover a portion of the surface **210** of the first housing **200**.

(42) FIG. **5** illustrates a second isometric view of the second assembly **400** of FIG. **4** including the first assembly **300** of FIG. **3** and the first electrical interconnect **410** of FIG. **4**. In the example of FIG. **5**, the surface **430** of the first electrical interconnect **410** includes the third plurality of terminals **440** and a second example edge **510**. The second edge **510** is configured to be the result of the mechanical alignment of the first housing **200**, such that the surface **210** may be mechanically coupled to the surface **430** to determine the dimensions of the second edge **510**. The dimensions of the second edge **510** are based on the difference between the dimensions of the second opening **420** and the dimensions of the surface **430** of the first electrical interconnect **410**. The second edge **510** is configured to extend around the surface **210** of the first housing **200** to the edge of the surface **430** of the first electrical interconnect **410**. The second edge **510** is configured to include the third plurality of terminals **440**, such that the third plurality of terminals **440** may protrude out of the surface **430** of the first electrical interconnect **410**. Alternatively, the third plurality of terminals **440** may be flush with the surface **430** of the first electrical interconnect **410**.

(43) FIG. **6** illustrates a third isometric view of the second example assembly **400** of FIGS. **4** and **5**. As shown in the illustrated example, the third plurality of terminals **440** may be configured to extend around the second edge **510**. Alternatively, the third plurality of terminals **440** may be configured to extend around a portion of the second edge **510**.

(44) FIG. **7** illustrates a bottom view of the first electrical interconnect **410** of FIGS. **4** and **5**. As shown in the illustrated example, the first electrical interconnect **410** includes the second opening **420**, the surface **430**, the third plurality of terminals **440**, the fourth plurality of terminals **450**, the fifth plurality of terminals **460**, and example mechanical guides **730-745**. In the example of FIG. **7**, the third plurality of terminals **440** are exposed when the surfaces **150** and **210** are mechanically coupled to the surface **430**. Alternatively, the first electrical interconnect **410** may include one or more terminals configured to couple to one or more terminals of the first integrated circuit **100** of FIG. **1**. In the example of FIG. **7**, the second opening **420** includes the mechanical guides **730-745**. The mechanical guides **730-745** are configured to mechanically align the first electrical interconnect **410** to the first assembly **300** of FIG. **3**, such that the optical interface and electrical interface are aligned during the manufacturing process. Alternatively, the mechanical guides **730-745** may be configured to be included in the second opening **420**, such that the mechanical guides may mechanically align the first electrical interconnect **410** to an integrated circuit (e.g., the first integrated circuit **100** of FIG. **1**).

(45) FIG. **8** illustrates a bottom view of a second example electrical interconnect **800**. As shown in the illustrated example, the second electrical interconnect **800** includes an example surface **810**, a third example opening **820**, a sixth example plurality of terminals **830**, a seventh plurality of terminals **840**, an eighth example plurality of terminals **850**, a ninth example plurality of terminals **860**, a tenth example plurality of terminals **870**, and an eleventh plurality of terminals **880**. The first electrical interconnect **410** of FIGS. **4-6** may be the first electrical interconnect **700** or the second electrical interconnect **800**.

(46) In the example of FIG. **8**, the surface **810** of the second electrical interconnect **800** includes the third opening **820**, such that the optical interface of the first integrated circuit **100** may be aligned. Alternatively, the third opening **820** may be configured based on the optical and/or electrical interface of the integrated circuit (e.g., the first integrated circuit **100**) being packaged. The

plurality of terminals **850-780** are electrically coupled to the sixth plurality of terminals **830** and/or the seventh plurality of terminals **840**. The sixth plurality of terminals **830** are configured to be electrically coupled to the first plurality of terminals **120**. The seventh plurality of terminals **840** are electrically coupled to the second plurality of terminals **130**. Alternatively, the second electrical interconnect **800** may include one or more terminals configured to be electrically coupled to one or more terminals of the first integrated circuit **100**.

(47) FIG. **9** illustrates a cross-sectional view of an example assembly apparatus **900** to assemble the second assembly **400** of FIGS. **4-6**. The assembly apparatus **900** is configured to illustrate the method implemented to electrically couple the first electrical interconnect **410** to the first assembly **300** of FIG. **3**. As shown in the illustrated example, the assembly apparatus **900** includes the first assembly **300**, the first electrical interconnect **410**, a first example heat source **905**, a second example heat source **910**, a first example conductive film **915**, a second conductive film **920**, and a first example adhesive **930**. The conductive films **915** and **920** may be an anisotropic conductive film (e.g., solder paste, conductive adhesive, low temperature silver epoxy, etc.).

(48) As shown in the illustrated example, the first integrated circuit **100** is mechanically coupled to the first housing **200** by the first adhesive **930**, such that the surface **150** of the first integrated circuit **100** is aligned with the surface **210** of the first housing **200**. The first conductive film **915** is configured to be electrically coupled to the first plurality of terminals **120** and to the fourth plurality of terminals **450**. The second conductive film **920** is configured to be electrically coupled to the second plurality of terminals **130** and to the fifth plurality of terminals **460**. The conductive films **915** and **920** may be assembled to be electrically coupled, such that the surface **150** of the first integrated circuit **100** is electrically coupled to the surface **430** of the first electrical interconnect **410**. The conductive films **915** and **920** (e.g., solder paste, conductive adhesive, low temperature silver epoxy, etc.) may be configured to be mechanically coupled to the surfaces **210** and **430**, such that the surface **210** of the first housing **200** of FIG. **2** is mechanically coupled to the surface **430** of the first electrical interconnect **410**. Advantageously, the conductive films **915** and **920** enable the assembly of the second assembly **400** of FIG. **4** to be performed by applying heat to the components.

(49) As shown in the illustrated example, the first heat source **905** is configured to apply sufficient heat to an example surface **940**, such that the surfaces **150** and/or **210** are mechanically coupled to the surface **430** of the first electrical interconnect **410** by the first conductive film **915**. The amount of heat applied to the surface **940** is based on the material the conductive films **915** and **920** are manufactured from and/or limited by the amount of heat that may be applied to the first integrated circuit **100**. The second heat source **910** is configured to apply sufficient heat to the surface **940** of the first electrical interconnect **410**, such that the surfaces **150** and/or **210** are mechanically coupled to the surface **430** of the first electrical interconnect **410** by the second conductive film **920**. In the example of FIG. **9**, the heat sources **905** and **910** are configured to supply heat to an area on the surface **940** greater than the area of the plurality of terminals **120** and **130**. The first edge **230** is configured to enable the area of the heat source **905** and **910** to be greater, such that the pressure applied to the surface **940** may not damage the components. Advantageously, the first edge **230** enables simpler manufacturing of the second assembly **400** of FIG. **4** by enabling greater contact area of the heat source **905** and **910** with the surface **940**.

(50) As shown in the illustrated example, the first electrical interconnect **410** is comprised of a first example layer **945**, a second example layer **950** and a third layer **955**. The first layer **945** is configured separate the heat sources **905** and **910** from the second layer **950**. The first layer **945** may be manufactured as an example insulator. The second layer **950** is manufactured as a conductor that may be electrically coupled to the plurality of terminals **440**, **450**, and/or **460**. Alternatively, portions of the second layer **950** may be separated by an insulating material, such that the second layer **950** may be comprised of a plurality of conductive traces to individually be electrically coupled a terminal from the plurality of terminals **440**, **450**, and/or **460** to another

terminal from the plurality of terminals **440**, **450**, and/or **460**. The third layer **955** is manufactured as an example insulator including portions of conductive terminals to be electrically coupled to portions of the second layer **950**. Alternatively, the first electrical interconnect **410** may be any electrical interconnect (e.g., the second electrical interconnect **800** of FIG. 8).

(51) FIG. **10** illustrates a bottom view of a third assembly **1000** including the second assembly **400** of FIG. **4** and a second example integrated circuit **1010**. The second integrated circuit **1010** includes a fourth example opening **1020**. The second integrated circuit **1010** may be configured to include processing circuitry, memory, and/or additional circuitry to be electrically coupled to the first integrated circuit **100** of FIG. **1** through the third plurality of terminals **440**. Alternatively, the second integrated circuit **1010** may be a motherboard or PCB. The fourth opening **1020** is configured to fit the first housing **200** of FIG. **2**, such that the fourth opening **1020** may be around the first assembly **300** of FIG. **3**, with the first electrical interconnect **410** of FIG. **4** electrically coupled to the second integrated circuit **1010**. The surface **430** of the first electrical interconnect **410** extends above the top surface of the second integrated circuit **1010**, such that the third plurality of terminals **440** may be electrically coupled to the second integrated circuit **1010**. The third plurality of terminals **440** may be electrically coupled to the second integrated circuit **1010** using the method described in FIG. **9**.

(52) FIG. **11** illustrates a cross-sectional view of the second integrated circuit **1010** of FIG. **10** along line **14-14**. As shown in the illustrated example, the second integrated circuit **1010** includes the fourth opening **1020** and a surface **1100**. The fourth opening **1020** is manufactured to house the first housing **200** of FIG. **2**. The surface **1100** of the second integrated circuit **1010** may include a plurality of terminals to be electrically coupled to additional circuitry (e.g., an optical IC, a processor, a sensor, etc.).

(53) FIG. **12** illustrates a second cross-sectional view of the second integrated circuit **1010** of FIG. **10** along line **14-14** including an example adhesive. As shown in the illustrated example, the second integrated circuit **1010** includes the fourth opening **1020**, the surface **1100**, a third example conductive film **1210**, and a fourth example conductive film **1220**. The conductive films **1210** and **1220** (e.g., solder paste, conductive adhesive, low temperature silver epoxy, etc.) may be electrically coupled to a plurality of terminals on the surface **1100** of the second integrated circuit **1010**.

(54) FIG. **13** illustrates a cross-sectional view of the third assembly **1000** of FIG. **10** along line **14-14**. As shown in the illustrated example, the second assembly **400** is mechanically aligned, such that the first housing **200** may be housed in the fourth opening **1020**. Advantageously, the mechanical alignment of the first housing **200** is configured to mechanically align the third plurality of terminals **440**, such that the third plurality of terminals may be electrically coupled to the conductive films **1210** and **1220**.

(55) FIG. **14** illustrates a cross-sectional view taken along line **10-10** of FIG. **10**. As shown in the illustrated example, the third assembly **1000** including the second assembly **400** of FIG. **4**, the second integrated circuit **1010**, and the surface **1100**. In the example of FIG. **14**, the first housing **200** is surrounded by the fourth opening **1020**, such that the surfaces **150** and **210** of the first assembly **300** of FIG. **3** are aligned with the surface **1100** of the second integrated circuit **1010**. The surface **430** of the first electrical interconnect **410** is mechanically coupled to the surface **1100** of the second integrated circuit **1010**. The third plurality of terminals **440** may be electrically coupled to a plurality of terminals on the surface **1100**. The plurality of terminals on the surface **1100** may be configured to be electrically coupled to the conductive films **1210** and **1220**. The plurality of terminals on the surface **1100** may be configured to be electrically coupled to a portion of the third plurality of terminals **440**. Advantageously, the fourth opening **1020** enables the second assembly **400** of FIG. **4** to be mounted within the second integrated circuit **1010** minimizing the additional area added by the second assembly **400** of FIG. **4**.

(56) FIG. **15** is a flowchart representative of an example method to manufacture the third assembly

of FIGS. **9** and **10**. The example method is adapted to manufacture any type of integrated circuit packaging including a first integrated circuit (e.g., the first integrated circuit **100** of FIG. **1**), an intermediate housing (e.g., the first housing **200** of FIG. **2**), and an electrical interconnect (e.g., the first electrical interconnect **410** of FIG. **4**, the second electrical interconnect **800** of FIG. **8**, etc.) to a second integrated circuit (e.g., the second integrated circuit **1010** of FIG. **10**).

(57) In the example of FIG. **15**, the process begins at block **1505** with mounting a first chip into an intermediate structural carrier. For example, mounting the first integrated circuit **100** of FIG. **1** into the first housing **200** of FIG. **2** to assemble the first assembly **300** of FIG. **3**. In some methods of manufacturing a layer of adhesive may be added between the first integrated circuit **100** and the first housing **200**. Alternatively, the first integrated circuit **100** and the first housing **200** may be manufactured such that no adhesive is required, such that the first integrated circuit **100** is housed by the first housing **200**. At block **1505**, the first integrated circuit **100** is mechanically coupled to the first housing **200**. The process proceeds to block **1510**.

(58) In the example of FIG. **15**, at block **1510**, an electrical interconnect is mounted to the intermediate structural carrier and/or to the first chip. For example, the first electrical interconnect **410** of FIGS. **4-6** is electrically coupled to the first integrated circuit **100**, as illustrated in the assembly apparatus **900** of FIG. **9**. The process of block **1510** may include the assembly apparatus **900** of FIG. **9**. The process of block **1510** may include the heat sources **905** and **910** of FIG. **9**, such that the heat sources **905** and **910** of FIG. **9** may heat the first electrical interconnect **410** of FIG. **4** to electrically couple the plurality of terminals **120** and **130** of FIG. **1** to the plurality of terminals **450** and **460** of FIG. **4**. The process of block **1510** is configured to supply enough heat to the first electrical interconnect **410** of FIG. **4**, such that the conductive films **915** and **920** form electrical bonds without providing sufficient heat to damage the first integrated circuit **100** of FIG. **1**. The process described in block **1510** is illustrated by the assembly apparatus **900**. Alternatively, the electrical interconnect may be mechanically coupled to the intermediate structural carrier by a layer of adhesive. The process proceeds to block **1515**.

(59) In the example of FIG. **15**, at block **1515** the electrical interconnect is mounted to a second chip. The assembly from block **1510** is mounted to the second chip, such that a plurality of terminals on the electrical interconnect may be electrically coupled to a corresponding plurality of terminals on the second chip. Block **1515** may result in the third assembly **1000** of FIG. **10**. Thereafter, the example method of FIG. **15** ends.

(60) Although an example method is described with reference to the flowchart illustrated in FIG. **15**, many other methods of manufacturing the example third assembly **1000** of FIGS. **9** and **10**. For example, the order of execution of the blocks may be changed, and/or some of the blocks described may be changed, eliminated, or combined. Similarly, additional operations may be included in the manufacturing process before, in between, or after the blocks shown in the illustrated example.

(61) FIG. **16** illustrates a top view of a fourth example assembly **1600** including the first assembly **300** of FIG. **3** and a third example electrical interconnect **1610**. The third electrical interconnect **1610** may be a flexible electrical interconnect. The third electrical interconnect **1610** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.) and insulator, designed to electrically couple a plurality of terminals to another plurality of terminals. In the example of the FIG. **16**, the third electrical interconnect **1610** includes a thirteenth plurality of terminals **1620** and a fourteenth plurality of terminals **1630**. The thirteenth plurality of terminals **1620** may be electrically coupled to the fourteenth plurality of terminals **1630**. In the example of FIG. **16**, the third electrical interconnect **1610** is electrically coupled to the first plurality of terminals **120** of FIG. **1** (not pictured in FIG. **16** for clarity). Alternatively, the third electrical interconnect **1610** may be electrically coupled to the second plurality of terminals **130** and/or the third electrical interconnect **1610** may be replicated to be electrically coupled to the second plurality of terminals **130**. The assembly apparatus **900** of FIG. **9** may be configured to electrically couple the third electrical interconnect **1610** to the first assembly **300**, such that the heat provided

by the heat sources **905** and **910** electrically couples the plurality of terminals **120** and **130** to **1620** without damaging the first integrated circuit **100** of FIG. **1**.

(62) FIG. **17** illustrates an isometric view of the fourth assembly **1600** including the first assembly **300** of FIG. **3** and the third electrical interconnect **1610** of FIG. **16**. In the example of FIG. **16**, the third electrical interconnect **1610** includes the thirteenth plurality of terminals **1620**, the fourteenth plurality of terminals **1630**, and an example surface **1700**. The surface **1700** of the third electrical interconnect **1610** may include the thirteenth plurality of terminals **1620** and the fourteenth plurality of terminals **1630**. The plurality of terminals **1620** and **1630** are configured to be electrically coupled by either surface of the third electrical interconnect **1610**, such that the surface **1700** of the third electrical interconnect **1610** may be electrically coupled to the first plurality of terminals **120** of FIG. **1** (not pictured in FIG. **17** for clarity). The surface **1700** is mechanically coupled to the surface **150** of the first integrated circuit **100** of FIG. **1** and/or the surface **210** of the first housing **200** of FIG. **2** (not pictured in FIG. **17** for clarity). The surface **1700** may be configured to include the thirteenth plurality of terminals **1620**, such that the thirteenth plurality of terminals **1620** may be electrically coupled to the first plurality of terminals **120** of FIG. **1**. Alternatively, the surface **1700** may include the fourteenth plurality of terminals **1630**, such that the third electrical interconnect **1610** may be electrically coupled to a third integrated circuit (not illustrated).

(63) FIG. **18** illustrates a side view of a fifth example assembly **1800** including the fourth assembly **1600** of FIGS. **12** and **13**, and a fourth example electrical interconnect **1810**. As shown in the illustrated example, the fifth assembly **1800** includes the fourth assembly **1600** of FIGS. **12** and **13**, the fourth electrical interconnect **1810**, and a third example integrated circuit **1820**. The third integrated circuit **1820** may be configured to include processing circuitry, memory, and/or additional circuitry to be electrically coupled to the first integrated circuit **100** of FIG. **1** through a plurality of terminals on the surface **1700** of the third electrical interconnect **1610**. Alternatively, the second integrated circuit **1010** may be a motherboard or PCB. The first assembly **300** of FIG. **3** may be mechanically aligned to a surface **1840** of the third integrated circuit **1820**, such that the third electrical interconnect **1610** and fourth electrical interconnect **1810** are aligned to be electrically coupled to the third integrated circuit **1820**. The first assembly **300** may additionally be mechanically coupled to the surface **1840** with the adhesive (e.g., silicon, epoxy, acrylic, etc.) between the first assembly **300** and the surface **1840**. The fourth electrical interconnect **1810** is manufactured similar (preferably identical) to that of the third electrical interconnect **1610** of FIGS. **12** and **13**.

(64) In the example of FIG. **18**, the fourth electrical interconnect **1810** includes an example surface **1830**. In the example of FIG. **18**, the electrical interconnects **1610** and **1810** are flexible electrical interconnects. Alternatively, the electrical interconnects **1610** and **1810** may be any electrical interconnect, such that the first plurality of terminals **120** of FIG. **1** and the second plurality of terminals **130** of FIG. **1** may be electrically coupled to the surface **1840**. In the example of FIG. **18**, the third electrical interconnect **1610** is electrically coupled to the first plurality of terminals **120** of FIG. **1** (not pictured in FIG. **18** for clarity) and a corresponding plurality of terminals on the surface **1840**. The third electrical interconnect **1610** may additionally be mechanically coupled to the surface **210** of the first housing **200** of FIG. **2** and the surface **1840**. In the example of FIG. **18**, the fourth electrical interconnect **1810** is electrically coupled to the second plurality of terminals **130** of FIG. **1** (not pictured in FIG. **18** for clarity) and a corresponding plurality of terminals on the surface **1840**. The fourth electrical interconnect **1810** may additionally be mechanically coupled to the surface **210** of the first housing **200** and the surface **1840**.

(65) FIG. **19** is a flowchart representative of an example method to manufacture the fifth assembly **1800** of FIG. **18**. The example method is adapted to manufacture any type of integrated circuit packaging including a first example integrated circuit (e.g., the first integrated circuit **100**), an intermediate structural carrier (e.g., the first housing **200** of FIG. **2**), and an example flexible

electrical interconnect (e.g., the third electrical interconnect **1610** of FIGS. **12-14**, the fourth electrical interconnect **1810** of FIG. **18**, etc.) to a second integrated circuit (e.g., the third integrated circuit **1820** of FIG. **18**).

(66) In the example of FIG. **19**, the process begins at block **1905** with mounting a first chip into an intermediate structural carrier. The first chip may be an integrated circuit (e.g., the first integrated circuit **100** of FIG. **1**, etc.). For example, mounting the first integrated circuit **100** of FIG. **1** into the first housing **200** of FIG. **2**, such that the first assembly **300** of FIG. **3** is the result. Additionally, adhesive (e.g., silicon, epoxy, acrylic, etc.) may be applied to mechanically couple the first chip and the intermediate structural carrier. The process proceeds to block **1910**.

(67) In the example of FIG. **19**, at block **1910**, a flexible interconnect is mounted to the first chip. For example, the electrical interconnect **1610** of FIG. **12-14** may be electrically and/or mechanically coupled to the first integrated circuit **100** using the assembly apparatus **900** of FIG. **9** configured to electrically couple the thirteenth plurality of terminals **1620** of FIGS. **12** and **13** to the first plurality of terminals **120** of FIG. **1**. An additional block may be implemented to mount a second flexible interconnect to the first chip using the same method described herein for block **1910**. The process proceeds to block **1915**.

(68) In the example of FIG. **19**, at block **1915**, the intermediate structural carrier is mounted to a second chip. For example, the first assembly **300** is mechanically coupled to the third integrated circuit **1820** of FIG. **18**. Additionally, adhesive (e.g., silicon, epoxy, acrylic, etc.) may be applied between the intermediate structural carrier and the second chip. The process proceeds to block **1920**.

(69) In the example of FIG. **19**, at block **1920**, the flexible interconnect is electrically and/or mechanically coupled to the second chip. For example, the third electrical interconnect **1610** is electrically coupled to the third integrated circuit **1820**. The fourteenth plurality of terminals **1630** of FIGS. **12** and **13** may be electrically coupled to a corresponding plurality of terminals on the surface **1840** of the third integrated circuit **1820** of FIG. **18**. An additional block may be implemented to electrically and/or mechanically couple a second flexible interconnect to the second chip, such as illustrated by the fourth electrical interconnect **1810** of FIG. **18**. Thereafter, the example method of FIG. **19** ends.

(70) Although an example method is described with reference to the flowchart illustrated in FIG. **19**, many other methods of manufacturing the fifth assembly **1800** of FIG. **18**. For example, the order of execution of the blocks may be changed, and/or some of the blocks described may be changed, eliminated, or combined. Similarly, additional operations may be included in the manufacturing process before, in between, or after the blocks shown in the illustrated example.

(71) FIG. **20** illustrates a side view of a sixth example assembly **2000** including the first integrated circuit **100** of FIG. **1**, a plurality of electrical interconnects **2005** and **2010**, and a plurality of stand-offs **2015** and **2020**. As shown in the illustrated example, the sixth assembly **2000** includes the first integrated circuit **100**, a fifth example electrical interconnect **2005**, a sixth example electrical interconnect **2010**, a first example stand-off **2015**, a second example stand-off **2020**, and a seventh example electrical interconnect **2025**. Alternatively, the first integrated circuit **100** may be any integrated circuit. Alternatively, the seventh electrical interconnect **2025** may be an integrated circuit (e.g., the third integrated circuit **1820** of FIG. **18**). Alternatively, the sixth assembly **2000** may include any number of electrical interconnects (e.g., the fifth electrical interconnect **2005** and the sixth electrical interconnect **2010**) and stand-offs (e.g., the first stand-off **2015** and the second stand-off **2020**).

(72) In the example of FIG. **20**, the first integrated circuit **100** includes the surface **150**, the first plurality of terminals **120**, the second plurality of terminals **130**, the optical interface **140**, a fifteenth example plurality of terminals **2030**, and a sixteenth example plurality of terminals **2035**. Alternatively, the first integrated circuit **100** may be implemented with the first plurality of terminals **120** and the second plurality of terminals **130**, as illustrated in FIGS. **1-3**. Alternatively,

the first integrated circuit **100** of FIG. **1** may be implemented without the optical interface **140**.

(73) In the example of FIG. **20**, the fifth electrical interconnect **2005** includes an example surface **2045**. The surface **2045** of the fifth electrical interconnect **2005** may be electrically and/or mechanically coupled to the surface **150** of the first integrated circuit **100**. The fifth electrical interconnect **2005** may include a plurality of terminals on the surface **2045** (not pictured for clarity) to be electrically coupled to the first plurality of terminals **120** and/or the fifteenth plurality of terminals **2030**. The fifth electrical interconnect **2005** is configured to enable the optical interface **140** to be mechanically aligned.

(74) In the example of FIG. **20**, the sixth electrical interconnect **2010** includes an example surface **2050**. The surface **2050** of the sixth electrical interconnect **2010** may be electrically and/or mechanically coupled to the surface **150** of the first integrated circuit **100**. The sixth electrical interconnect **2010** may include a plurality of terminals on the surface **2050** (not pictured for clarity) to be electrically coupled to the second plurality of terminals **130** and/or the sixteenth plurality of terminals **2035**. The sixth electrical interconnect **2010** is configured to enable the optical interface **140** to be mechanically aligned.

(75) In the example of FIG. **20**, the first stand-off **2015** includes an example surface **2055** and an example surface **2060**. The surface **2055** of the first stand-off **2015** is mechanically coupled to the surface **2045** of the fifth electrical interconnect **2005**. Advantageously, the first stand-off **2015** enables the assembly apparatus **900** of FIG. **9** to be configured to electrically couple the fifth electrical interconnect **2005** to the first plurality of terminals **120** and/or the fifteenth plurality of terminals **2030**. Advantageously, the first stand-off **2015** increases the durability of the coupling of the fifth electrical interconnect **2005** to the first integrated circuit **100** of FIG. **1**.

(76) In the example of FIG. **20**, the second stand-off **2020** includes an example surface **2065** and an example surface **2070**. The surface **2065** of the second stand-off **2020** is mechanically coupled to the surface **2050** of the sixth electrical interconnect **2010**. Advantageously, the second stand-off **2020** enables the assembly apparatus **900** of FIG. **9** to be configured to electrically couple the sixth electrical interconnect **2010** to the second plurality of terminals **130** and/or the sixteenth plurality of terminals **2035**. Advantageously, the second stand-off **2020** increases the durability of the coupling of the sixth electrical interconnect **2010** to the first integrated circuit **100** of FIG. **1**.

(77) In the example of FIG. **20**, the seventh electrical interconnect **2025** includes an example surface **2075**, an example surface **2080**, a seventeenth example plurality of terminals **2085**, an example electrical interconnect layer **2090**, an eighteenth example plurality of terminals **2092**, a nineteenth example plurality of terminals **2094**, a twentieth example plurality of terminals **2096**, and a twenty-first plurality of terminals **2098**. The first integrated circuit **100** of FIG. **1** may be mechanically coupled to the surface **2075** of the seventh electrical interconnect **2025** by a second example adhesive layer **2088** (e.g., silicon, epoxy, acrylic, etc.). Alternatively, the first integrated circuit **100** may be mechanically coupled to the surface **2075** without the second adhesive layer **2088**. The surfaces **2045** and **2050** may be electrically and/or mechanically coupled to the surface **2075** of the seventh electrical interconnect **2025**. The surfaces **2060** and **2070** may be mechanically coupled to the surface **2075** of the seventh electrical interconnect **2025**. The surface **2080** of the seventh electrical interconnect **2025** includes the seventeenth plurality of terminals **2085**. The seventeenth plurality of terminals **2085** may be electrically coupled to a plurality of terminals on the surface **2075** of the seventh electrical interconnect **2025**. The seventh electrical interconnect **2025** may include the electrical interconnect layer **2090** to electrically couple the plurality of terminals **2092-2098** to the seventeenth plurality of terminals **2085**. The electrical interconnect layer **2090** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.) and insulator, designed to electrically couple a plurality of terminals to another plurality of terminals.

(78) Alternatively, the seventh electrical interconnect **2025** may be an integrated circuit (e.g., the third integrated circuit **1820** of FIG. **18**). Advantageously, the seventh electrical interconnect **2025**

enables the first integrated circuit **100** to be electrically coupled to an integrated circuit, such that the electrical and/or optical interface of the first integrated circuit **100** may be aligned.

(79) FIG. **21** is a flowchart representative of an example method to manufacture the sixth assembly **2000** of FIG. **20**. The example method is adapted to manufacture any type of integrated circuit packaging including a chip (e.g., the first integrated circuit **100**), a stand-off (e.g., the first stand-off **2015** of FIG. **20** and the second stand-off **2020** of FIG. **20**), and an example flexible electrical interconnect (e.g., the third electrical interconnect **1610** of FIGS. **12-14**, the fourth electrical interconnect **1810** of FIG. **18**, the fifth electrical interconnect **2005** of FIG. **20**, the sixth electrical interconnect **2010** of FIG. **20**, etc.) to an electrical interconnect (e.g., the seventh electrical interconnect **2025** of FIG. **20**). Alternatively, the electrical interconnect may be an integrated circuit (e.g., the third integrated circuit **1820** of FIG. **18**).

(80) In the example of FIG. **21**, the process begins at block **2110** with mounting a chip onto an electrical interconnect. For example, mounting the first integrated circuit **100** of FIG. **1** onto the seventh electrical interconnect **2025** of FIG. **20**. An adhesive layer (e.g., the second adhesive layer **2088** of FIG. **20**) may be applied to mechanically couple the chip to the electrical interconnect. The process proceeds to block **2120**.

(81) In the example of FIG. **21**, at block **2120**, a stand-off is mounted to the electrical interconnect. For example, the first stand-off **2015** of FIG. **20** is mechanically coupled to the seventh electrical interconnect **2025**. An additional block may be added to mount an additional stand-off to the electrical interconnect, such as illustrated in the sixth assembly **2000** of FIG. **20** by the sixth electrical interconnect **2010** of FIG. **20** and the second stand-off **2020** of FIG. **20**. The process proceeds to block **2130**.

(82) In the example of FIG. **21**, at block **2130**, the flexible interconnect is mounted to the electrical interconnect. For example, the fifth electrical interconnect **2005** of FIG. **20** may be electrically and/or mechanically coupled to the seventh electrical interconnect **2025**. An additional block may be added to mount an additional flexible interconnect to the electrical interconnect, such as the sixth electrical interconnect **2010** electrically and/or mechanically coupled to the seventh electrical interconnect **2025**. The process proceeds to block **2140**.

(83) In the example of FIG. **21**, at block **2140**, the flexible interconnect is mounted to the chip. The chip of the process at block **2140** may be an integrated circuit (e.g., the third integrated circuit **1820** of FIG. **18**, the seventh electrical interconnect **2025** of FIG. **20**, etc.). For example, the fifth electrical interconnect **2005** of FIG. **20** may be electrically and/or mechanically coupled to the first integrated circuit **100**. The block **2140** may be completed by the assembly apparatus **900** of FIG. **9** configured to electrically and/or mechanically couple the flexible interconnects to the first integrated circuit **100**. An additional block may be added to mount an additional flexible interconnect to the chip, such as the sixth electrical interconnect **2010** is electrically and/or mechanically coupled to the first integrated circuit **100**. Thereafter, the example method of FIG. **19** ends.

(84) Although an example method is described with reference to the flowchart illustrated in FIG. **21**, many other methods of manufacturing the sixth assembly **2000**. For example, the order of execution of the blocks may be changed, and/or some of the blocks described may be changed, eliminated, or combined. Similarly, additional operations may be included in the manufacturing process before, in between, or after the blocks shown in the illustrated example.

(85) FIG. **22** is a side view of a seventh example assembly **2200**. As shown in the illustrated example, the seventh assembly **2200** includes the first integrated circuit **100**, an eighth example electrical interconnect **2205**, a ninth example electrical interconnect **2210**, and an eleventh example electrical interconnect **2215**. Alternatively, the first integrated circuit **100** may be any integrated circuit. Alternatively, the eleventh electrical interconnect **2215** may be an integrated circuit (e.g., the third integrated circuit **1820** of FIG. **18**, the seventh electrical interconnect **2025** of FIG. **20**, etc.). Alternatively, the seventh assembly **2200** may include any number of electrical interconnects

(e.g., the fifth electrical interconnect **2005**, and the sixth electrical interconnect **2010**).

(86) In the example of FIG. 22, the first integrated circuit **100** includes the surface **150**, the first plurality of terminals **120**, the second plurality of terminals **130**, the optical interface **140**, the fifteenth plurality of terminals **2030**, and the sixteenth plurality of terminals **2035**. Alternatively, the first integrated circuit **100** may be implemented with the first plurality of terminals **120** and the second plurality of terminals **130**, as illustrated in FIGS. 1-3. Alternatively, the first integrated circuit **100** of FIG. 1 may be implemented without the optical interface **140**.

(87) In the example of FIG. 22, the eighth electrical interconnect **2205** includes an example surface **2220**. The surface **2220** of the eighth electrical interconnect **2205** may be electrically and/or mechanically coupled to the surface **150** of the first integrated circuit **100**. The eighth electrical interconnect **2205** may include a plurality of terminals on the surface **2220** (not pictured for clarity) to be electrically coupled to the first plurality of terminals **120** and/or the fifteenth plurality of terminals **2030**. The eighth electrical interconnect **2205** is configured to enable the optical interface **140** to be mechanically aligned. The eighth electrical interconnect **2205** is electrically coupled to the plurality of terminals **120**, **2030**, and **2245-2250**, such that the eighth electrical interconnect **2205** may bend above the surface **150** of the optical IC **110**. Advantageously, the eighth electrical interconnect **2205** is configured to bend to modify the durability of the seventh assembly **2200** and enable multiple manufacturing methods.

(88) In the example of FIG. 22, the ninth electrical interconnect **2210** includes an example surface **2225**. The surface **2225** of the ninth electrical interconnect **2210** may be electrically and/or mechanically coupled to the surface **150** of the first integrated circuit **100**. The ninth electrical interconnect **2210** may include a plurality of terminals on the surface **2225** (not pictured for clarity) to be electrically coupled to the second plurality of terminals **130** and/or the sixteenth plurality of terminals **2035**. The ninth electrical interconnect **2210** is configured to enable the optical interface **140** to be mechanically aligned. The ninth electrical interconnect **2210** is electrically coupled to the plurality of terminals **130**, **2035**, and **2255-2260**, such that the ninth electrical interconnect **2210** may bend above the surface **150** of the optical IC **110**. Advantageously, the ninth electrical interconnect **2210** may be configured to bend to modify the durability of the seventh assembly **2200** and enable multiple manufacturing methods.

(89) In the example of FIG. 22, the eleventh electrical interconnect **2215** includes an example surface **2230**, an example surface **2235**, a twenty-second example plurality of terminals **2240**, a twenty-third example plurality of terminals **2245**, a twenty-fourth example plurality of terminals **2250**, a twenty-fifth example plurality of terminals **2255**, a twenty-sixth example plurality of terminals **2260**, and a second example electrical interconnect layer **2265**. The first integrated circuit **100** of FIG. 1 may be mechanically coupled to the surface **2230** of the eleventh electrical interconnect **2215** by a third example adhesive layer **2270** (e.g., silicon, epoxy, acrylic, etc.). Alternatively, the first integrated circuit **100** may be mechanically coupled to the surface **2230** without the third adhesive layer **2270**. The surfaces **2220** and **2225** may be electrically and/or mechanically coupled to the surface **2230** of the eleventh electrical interconnect **2215**. The surface **2235** of the eleventh electrical interconnect **2215** includes the twenty-second plurality of terminals **2240**. The twenty-second plurality of terminals **2240** may be electrically coupled to a plurality of terminals on the surface **2230** of the eleventh electrical interconnect **2215**. The eleventh electrical interconnect **2215** may include the second electrical interconnect layer **2265** to electrically couple the plurality of terminals **2245-2260** to the twenty-second plurality of terminals **2240**. The second electrical interconnect layer **2265** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.) and insulator, designed to electrically couple a plurality of terminals to another plurality of terminals.

(90) Alternatively, the eleventh electrical interconnect **2215** may be an integrated circuit (e.g., the third integrated circuit **1820** of FIG. 18). Advantageously, the eleventh electrical interconnect **2215** enables the first integrated circuit **100** to be electrically coupled to an integrated circuit, such that

the electrical and/or optical interface of the first integrated circuit **100** may be aligned.

(91) FIG. **23** is a side view of an eighth example assembly **2300**. As shown in the illustrated example, the eighth assembly **2300** includes the first assembly **300**, the eighth electrical interconnect **2205**, the ninth electrical interconnect **2210**, and the eleventh example electrical interconnect **2215**.

(92) In the example of FIG. **23**, the eighth electrical interconnect **2205** is electrically coupled to the plurality of terminals **120**, **2030**, and **2245-2250**, such that the eighth electrical interconnect **2205** may bend away from the optical IC **110**. Advantageously, the eighth electrical interconnect **2205** is configured to bend to modify the durability of the eighth assembly **2300** and enable multiple manufacturing methods.

(93) In the example of FIG. **23**, the ninth electrical interconnect **2210** is electrically coupled to the plurality of terminals **130**, **2035**, and **2255-2260**, such that the ninth electrical interconnect **2210** may bend away from the optical IC **110**. Advantageously, the ninth electrical interconnect **2210** may be configured to bend to modify the durability of the eighth assembly **2300** and enable multiple manufacturing methods.

(94) FIG. **24** is a side view of a ninth example assembly **2400**. As shown in the illustrated example, the ninth assembly **2400** includes the first integrated circuit **100**, the eleventh example electrical interconnect **2215**, a twelfth example electrical interconnect **2405**, and a thirteenth example electrical interconnect **2410**. Alternatively, the seventh assembly **2200** may include any number of electrical interconnects (e.g., the fifth electrical interconnect **2005** and the sixth electrical interconnect **2010**).

(95) In the example of FIG. **24**, the twelfth electrical interconnect **2405** includes a first example surface **2415** and a second example surface **2420**. The first surface **2415** of the twelfth electrical interconnect **2405** may be electrically and/or mechanically coupled to the surface **150** of the first integrated circuit **100**. The twelfth electrical interconnect **2405** may include a plurality of terminals on the first surface **2415** (not pictured for clarity) and the second surface **2420** (not pictured for clarity) to be electrically coupled to the first plurality of terminals **120** and/or the fifteenth plurality of terminals **2030**. The twelfth electrical interconnect **2405** is configured to enable the optical interface **140** to be mechanically aligned. The twelfth electrical interconnect **2405** is electrically coupled to the plurality of terminals **120** and/or **2030** by a plurality of terminals on the first surface **2415** (not pictured for clarity) and **2245-2250** on the second surface **2420** (not pictured for clarity), such that the twelfth electrical interconnect **2405** may bend away from the optical IC **110**.

Advantageously, the twelfth electrical interconnect **2405** is configured to bend to modify the durability of the ninth assembly **2400** and enable multiple manufacturing methods.

(96) In the example of FIG. **24**, the thirteenth electrical interconnect **2410** includes a first example surface **2425** and a second example surface **2430**. The first surface **2425** of the thirteenth electrical interconnect **2410** may be electrically and/or mechanically coupled to the surface **150** of the first integrated circuit **100**. The thirteenth electrical interconnect **2410** may include a plurality of terminals on the first surface **2425** (not pictured for clarity) and the second surface **2430** (not pictured for clarity) to be electrically coupled to the second plurality of terminals **130** and/or the sixteenth plurality of terminals **2035**. The thirteenth electrical interconnect **2410** is configured to enable the optical interface **140** to be mechanically aligned. The thirteenth electrical interconnect **2410** is electrically coupled to the plurality of terminals **130** and/or **2035** by a plurality of terminals on the first surface **2425** (not pictured for clarity), and **2255-2260** by a plurality of terminals on the second surface **2430** (not pictured for clarity), such that the thirteenth electrical interconnect **2410** may bend away from the optical IC **110**. Advantageously, the thirteenth electrical interconnect **2410** may be configured to bend to modify the durability of the ninth assembly **2400** and enable multiple manufacturing methods.

(97) FIG. **25** is a side view of a tenth example assembly **2500**. As shown in the illustrated example, the tenth assembly **2500** includes the first integrated circuit **100**, the eleventh example electrical

interconnect **2215**, a fourteenth example electrical interconnect **2505**, a fifteenth example electrical interconnect **2510**, and a second housing **2535**. Alternatively, the tenth assembly **2500** may include any number of electrical interconnects (e.g., the fifth electrical interconnect **2005** and the sixth electrical interconnect **2010**).

(98) In the example of FIG. 25, the fourteenth electrical interconnect **2505** includes a first example surface **2515** and a second example surface **2520**. The first surface **2515** of the fourteenth electrical interconnect **2505** may be electrically and/or mechanically coupled to the surface **150** of the first integrated circuit **100**. The fourteenth electrical interconnect **2505** may include a plurality of terminals on the first surface **2515** (not pictured for clarity) and the second surface **2520** (not pictured for clarity) to be electrically coupled to the first plurality of terminals **120** and/or the fifteenth plurality of terminals **2030**. The fourteenth electrical interconnect **2505** is configured to enable the optical interface **140** to be mechanically aligned. The fourteenth electrical interconnect **2505** is electrically coupled to the plurality of terminals **120** and/or **2030** by a plurality of terminals on the first surface **2515** (not pictured for clarity) and **2245-2250** on the second surface **2520** (not pictured for clarity), such that the fourteenth electrical interconnect **2505** may bend away from the optical IC **110**. Advantageously, the fourteenth electrical interconnect **2505** is configured to bend to modify the durability of the tenth assembly **2500** and enable multiple manufacturing methods.

(99) In the example of FIG. 25, the fifteenth electrical interconnect **2510** includes a first example surface **2525** and a second example surface **2530**. The first surface **2525** of the fifteenth electrical interconnect **2510** may be electrically and/or mechanically coupled to the surface **150** of the first integrated circuit **100**. The fifteenth electrical interconnect **2510** may include a plurality of terminals on the first surface **2525** (not pictured for clarity) and the second surface **2530** (not pictured for clarity) to be electrically coupled to the second plurality of terminals **130** and/or the sixteenth plurality of terminals **2035**. The fifteenth electrical interconnect **2510** is configured to enable the optical interface **140** to be mechanically aligned. The fifteenth electrical interconnect **2510** is electrically coupled to the plurality of terminals **130** and/or **2035** by a plurality of terminals on the first surface **2525** (not pictured for clarity), and **2255-2260** by a plurality of terminals on the second surface **2530** (not pictured for clarity), such that the fifteenth electrical interconnect **2510** may bend away from the optical IC **110**. Advantageously, the fifteenth electrical interconnect **2510** may be configured to bend to modify the durability of the tenth assembly **2500** and enable multiple manufacturing methods.

(100) In the example of FIG. 25, the second housing **2535** is configured to mechanically couple the first integrated circuit **100** to the eleventh electrical interconnect **2215**. The second housing **2535** is mechanically aligned, such that the optical interface **140** may be aligned with the desired location of the optical output. The second housing **2535** is manufactured as an insulator (e.g., epoxy material, etc.). The second housing is manufactured to house a portion of the first assembly **300**, such that the electrical interconnects **2505** and **2510** may not be in contact with the second housing **2535**. Advantageously, the second housing **2535** enables multiple manufacturing methods to mechanically align and/or couple the first integrated circuit **100** to the eleventh electrical interconnect **2215**.

(101) FIG. 26 is a side view of an eleventh example assembly **2600**. As shown in the illustrated example, the eleventh example assembly **2600** includes the first integrated circuit **100**, the eighth electrical interconnect **2205**, the ninth electrical interconnect **2210**, a sixteenth example electrical interconnect **2605**, and an example interconnect layer **2610**. The interconnect layer **2610** may be manufactured from a conductive film (e.g., solder paste, conductive adhesive, low temperature silver epoxy, etc.), such that a plurality of terminals from the first integrated circuit **100** may be mechanically and/or electrically coupled to the sixteenth electrical interconnect **2605**.

(102) In the example of FIG. 26, the sixteenth electrical interconnect **2605** includes a first example surface **2615**, a second example surface **2620**, a twenty-seventh example plurality of terminals **2625**, a twenty-eighth example plurality of terminals **2630**, a twenty-ninth example plurality of

terminals **2635**, a thirtieth plurality of terminals **2640**, a thirty-first plurality of terminals **2645**, a thirty-second plurality of terminals **2650**, and a third electrical interconnect layer **2655**. The sixteenth electrical interconnect **2605** may include the third electrical interconnect layer **2655** to electrically couple the plurality of terminals **2630-2650** to the twenty-seventh plurality of terminals **2625**. The third electrical interconnect layer **2655** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.) and insulator, designed to electrically couple a plurality of terminals to another plurality of terminals. Advantageously, the sixteenth electrical interconnect **2605** enables the first integrated circuit **100** to be implemented as an integrated circuit with a plurality of terminals electrically and/or mechanically coupled directly to the sixteenth electrical interconnect **2605**.

(103) FIG. **27** is a cross-sectional view of a twelfth example assembly **2700**. As shown in the illustrated example, the twelfth example assembly **2700** includes the first integrated circuit **100**, the third adhesive layer **2270**, a seventeenth example electrical interconnect **2705**, an eighteenth example electrical interconnect **2710**, a first mechanical interconnect **2715**, a second mechanical interconnect **2720**, a nineteenth example electrical interconnect **2725**, and a twentieth example electrical interconnect **2730**. The mechanical interconnects **2715** and **2720** may be manufactured as an insulator (e.g., epoxy, etc.), such that each terminal of the plurality of terminals **120** and **130** may each be electrically coupled.

(104) In the example of FIG. **27**, the seventeenth electrical interconnect **2705** includes a first example surface **2735**, a second example surface **2740**, a thirty-third plurality of terminal **2745**, a thirty-fourth plurality of terminals **2750**, a thirty-fifth plurality of terminals **2755**, and a fourth electrical interconnect layer **2760**. The seventeenth electrical interconnect **2705** may include the fourth electrical interconnect layer **2760** to electrically couple the plurality of terminals **2745-2755** to the nineteenth electrical interconnect **2725**. The fourth electrical interconnect layer **2760** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.) and insulator, designed to electrically couple a plurality of terminals to another plurality of terminals or electrical interconnect.

(105) In the example of FIG. **27**, the eighteenth electrical interconnect **2710** includes a first example surface **2765**, a second example surface **2770**, a thirty-sixth plurality of terminal **2775**, a thirty-seventh plurality of terminals **2780**, a thirty-eighth plurality of terminals **2785**, and a fifth electrical interconnect layer **2790**. The eighteenth electrical interconnect **2710** may include the fifth electrical interconnect layer **2790** to electrically couple the plurality of terminals **2775-2785** to the twentieth electrical interconnect **2730**. The fifth electrical interconnect layer **2790** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.) and insulator, designed to electrically couple a plurality of terminals to another plurality of terminals or electrical interconnect.

(106) In the example of FIG. **27**, the first mechanical interconnect **2715** may be configured to encase the nineteenth electrical interconnect **2725**, such that the first plurality of terminals **120** may be electrically coupled to the seventeenth electrical interconnect **2705**. The second mechanical interconnect **2720** may be configured to encase the twentieth electrical interconnect **2730**, such that the second plurality of terminals **130** may be electrically coupled to the eighteenth electrical interconnect **2710**. Advantageously, the mechanical interconnects **2715** and **2720** may be configured to modify the durability of the twelfth assembly **2700**.

(107) FIG. **28** is a side view of a third example housing **2800**. As shown in the illustrated example, the third housing **2800** includes a first example surface **2805**, a second example surface **2810**, a third example surface **2815**, a fifth example opening **2820**, a thirty-ninth example plurality of terminals **2825**, a fortieth example plurality of terminals **2830**, a forty-first example plurality of terminals **2835**, a sixth example electrical interconnect layer **2840**, a forty-second example plurality of terminals **2845**, and a forty-third example plurality of terminals **2850**. The third housing **2800** may include the sixth electrical interconnect layer **2840** to electrically couple the plurality of

terminals **2830**, **2835**, **2845**, and **2850** to the thirty-ninth plurality of terminals **2825**. The sixth electrical interconnect layer **2840** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.) and insulator, designed to electrically couple a plurality of terminals to another plurality of terminals or electrical interconnect.

(108) In the example of FIG. **28**, the surfaces **2815** and **2810** may be configured to be flush within respect to each other. The fifth opening **2820** is configured to house an integrated circuit (e.g., the first integrated circuit **100** of FIG. **1**). Advantageously, the third housing **2800** may be manufactured using multiple methods.

(109) FIG. **29** is a side view of a thirteenth example assembly **2900**. As shown in the illustrated example, the thirteenth assembly **2900** includes the first integrated circuit **100**, the third adhesive layer **2270**, and the third housing **2800**. The first integrated circuit **100** may be mechanically coupled to the third housing **2800**, such that the third adhesive layer **2270** separates the first integrated circuit **100** and the third housing **2800**. The surface **150** of the first integrated circuit **100** may be flush to the surface **2810** and **2815** of the third housing **2800**.

(110) FIG. **30** is a side view of a fourteenth example assembly **3000** including the thirteenth assembly **2900**. As shown in the illustrated example, the fourteenth assembly **3000** includes the first integrated circuit **100**, the third housing **2800**, a twenty-first example electrical interconnect **3005**, and a twenty-second electrical interconnect **3010**.

(111) In the example of FIG. **30**, the twenty-first electrical interconnect **3005** includes a first example plurality of electrical traces **3015**, and a second example plurality of electrical traces **3020**. The first plurality of electrical traces **3015** may be configured to be electrically coupled to the fifteenth plurality of terminals **2030** and the fortieth plurality of terminals **2830**. The first plurality of electrical traces **3015** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.), designed to electrically couple a plurality of terminals to another plurality of terminals. The second plurality of electrical traces **3020** may be configured to be electrically coupled to the first plurality of terminals **120** and the forty-first plurality of terminals **2835**. The second plurality of electrical traces **3020** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.), designed to electrically couple a plurality of terminals to another plurality of terminals.

(112) In the example of FIG. **30**, the twenty-second electrical interconnect **3010** includes a third example plurality of electrical traces **3025**, and a fourth example plurality of electrical traces **3030**. The third plurality of electrical traces **3025** may be configured to be electrically coupled to the sixteenth plurality of terminals **2035** and the forty-third plurality of terminals **2850**. The third plurality of electrical traces **3025** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.), designed to electrically couple a plurality of terminals to another plurality of terminals. The fourth plurality of electrical traces **3030** may be configured to be electrically coupled to the second plurality of terminals **130** and the forty-second plurality of terminals **2845**. The fourth plurality of electrical traces **3030** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.), designed to electrically couple a plurality of terminals to another plurality of terminals.

(113) FIG. **31** is a side view of a fifteenth example assembly **3100** including the third housing **2800**. As shown in the illustrated example, the fifteenth assembly **3100** includes the first integrated circuit **100**, the third housing **2800**, a twenty-third example electrical interconnect **3105**, and a twenty-fourth electrical interconnect **3110**.

(114) In the example of FIG. **31**, the twenty-third electrical interconnect **3105** includes a fifth example plurality of electrical traces **3115**, and a sixth example plurality of electrical traces **3120**. The fifth plurality of electrical traces **3115** may be configured to be electrically coupled to the fifteenth plurality of terminals **2030** and the fortieth plurality of terminals **2830**. The fifth plurality of electrical traces **3115** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.), designed to electrically couple a plurality of terminals to another plurality

of terminals. The sixth plurality of electrical traces **3120** may be configured to be electrically coupled to the first plurality of terminals **120** and the forty-first plurality of terminals **2835**. The sixth plurality of electrical traces **3120** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.), designed to electrically couple a plurality of terminals to another plurality of terminals.

(115) In the example of FIG. **31**, the twenty-fourth electrical interconnect **3110** includes a seventh example plurality of electrical traces **3125**, and an eighth example plurality of electrical traces **3130**. The seventh plurality of electrical traces **3125** may be configured to be electrically coupled to the sixteenth plurality of terminals **2035** and the forty-third plurality of terminals **2850**. The seventh plurality of electrical traces **3125** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.), designed to electrically couple a plurality of terminals to another plurality of terminals. The eighth plurality of electrical traces **3130** may be configured to be electrically coupled to the second plurality of terminals **130** and the forty-second plurality of terminals **2845**. The eighth plurality of electrical traces **3130** may be manufactured out of any suitable conductive material (e.g., copper, nickel, silver, gold, etc.), designed to electrically couple a plurality of terminals to another plurality of terminals.

(116) In the example of FIG. **31**, the electrical interconnects **3105** and **3110** are configured to bend away from surface **2810** and **2815**. The electrical interconnects Advantageously, by bending the electrical interconnects **3105** and **3110**, the durability of the fifteenth assembly **3100** may be modified.

(117) Although certain example methods, apparatus and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all methods, apparatus and articles of manufacture fairly falling within the scope of the claims of this patent.

Claims

1. An apparatus comprising: an integrated circuit having a first surface and terminals on the first surface; a housing at least partially laterally surrounding the integrated circuit, the housing exposing the first surface; and an electrical interconnect having a second surface and an opening, the second surface electrically coupled to the terminals on the first surface of the integrated circuit, the second surface contacting the housing, the opening exposing the first surface.
2. The apparatus of claim 1, wherein the integrated circuit comprises an optical device.
3. The apparatus of claim 1, wherein the integrated circuit includes a conductive film between the first surface of the integrated circuit and the second surface of the electrical interconnect.
4. The apparatus of claim 1, wherein the integrated circuit includes a plurality of terminals on the first surface.
5. The apparatus of claim 1, wherein the housing includes a third surface contacting the second surface of the electrical interconnect.
6. The apparatus of claim 1, wherein the electrical interconnect includes the opening to expose an optical interface on the first surface of the integrated circuit.
7. The apparatus of claim 1, wherein the electrical interconnect includes a first plurality of terminals and a second plurality of terminals on the second surface.
8. The apparatus of claim 1, wherein the electrical interconnect includes a third surface, the opening configured to extend from the second surface to the third surface.
9. An apparatus comprising: a first integrated circuit having a first surface; a housing at least partially laterally surrounding the first integrated circuit, the housing having a second surface; an electrical interconnect having a third surface, the third surface electrically coupled to the first surface, the third surface contacting the second surface; and a second integrated circuit having a fourth surface, the fourth surface on the second surface, the fourth surface electrically coupled to

the third surface.

10. The apparatus of claim 9, wherein the first integrated circuit is an optical device.

11. The apparatus of claim 9, wherein the first integrated circuit includes a plurality of terminals on the first surface.

12. The apparatus of claim 9, wherein the housing includes a fifth surface contacting to the third surface.

13. The apparatus of claim 9, wherein the electrical interconnect includes a first plurality of terminals and a second plurality of terminals on the third surface.

14. The apparatus of claim 9, wherein the second integrated circuit includes a plurality of terminals on the fourth surface.

15. An apparatus comprising: an integrated circuit including a first surface and a second surface; a stand-off including a first surface and a second surface; and an electrical interconnect having a first surface and a second surface, the first surface contacting first surface of the stand-off, the second surface contacting the second surface of the stand-off, the electrical interconnect electrically coupled to the second surface of the integrated circuit and to the first surface of the electrical interconnect.

16. The apparatus of claim 15, wherein the integrated circuit is an optical device.

17. The apparatus of claim 15, wherein the first surface of the integrated circuit is coupled to the first surface of the electrical interconnect through adhesive.

18. The apparatus of claim 15, wherein the integrated circuit includes a plurality of terminals on the second surface of the integrated circuit.

19. The apparatus of claim 15, wherein the electrical interconnect includes a plurality of terminals on a third surface.

20. The apparatus of claim 15, wherein the electrical interconnect includes a fourth surface further including a plurality of terminals.
